

AI Career Intelligence Platform

AI Design Document

AI Architecture • Model Design • Prompt Engineering • Vector Database

Version: 1.0

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1. AI Architecture

The AI layer processes resumes, extracts skills, recommends jobs, predicts salaries, generates learning roadmaps, and powers the conversational career assistant. FastAPI orchestrates requests while AI services perform NLP and LLM tasks.

Pipeline

User → Resume Upload → Text Extraction → Skill Extraction → Embedding Generation → Job Matching → LLM Response → Frontend

2. Model Design

Component	Technology
Resume Parsing	Python PDF/Text Parser
Skill Extraction	spaCy NLP
Embeddings	Sentence Transformers
Job Matching	Cosine Similarity
Salary Prediction	Scikit-learn Regression
Career Assistant	LLM + LangChain

The architecture is modular so individual AI models can be upgraded without affecting the overall system.

3. Prompt Engineering

Prompt engineering ensures consistent and accurate responses from the LLM.

- Use structured system prompts.
- Provide resume context before user queries.
- Restrict responses to career-related guidance.
- Request structured JSON where downstream processing is required.
- Include safety instructions to reduce hallucinations.

4. Vector Database

Purpose	Description
Resume Embeddings	Semantic search across resumes
Job Embeddings	Semantic job matching
Knowledge Base	Career resources and FAQs
Similarity Search	Top-K nearest neighbour retrieval

Recommended Options: FAISS (development), ChromaDB (local), Pinecone or Weaviate (cloud deployment).

Retrieved documents are passed to the LLM using Retrieval-Augmented Generation (RAG) to improve response quality.

5. Conclusion

This document outlines the AI architecture for Version 1.0 of the AI Career Intelligence Platform. It defines the AI processing pipeline, model selection, prompt engineering strategy, and vector database integration needed for scalable intelligent career services.